Who Trades with Whom? Centrality and Communities in the Football Transfer Network (2010–2023)

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https://github.com/aarensberg/football-transfer-network-analysis

Abstract

We ask which football clubs occupy the most central positions in the transfer network and which communities of clubs mainly exchange players with each other. Starting from relational tables of 3.47 million transfer records scraped from Transfermarkt, we build a directed, simple, weighted graph of 12,009 clubs and 450,217 club-to-club links aggregating 920,304 transfers made between 2010 and 2023. Using Python and NetworkX, we compute exact connectivity, distances, centralities (degree, strength, closeness, betweenness, PageRank, eigenvector) and communities (Louvain, checked with Leiden, label propagation and a null model). The network is a small world (3–4 transfers separate two clubs; diameter 12 and radius 6 on the largest strongly connected component) and is highly local: 85% of transfers are domestic and Louvain finds 35 communities with a modularity of 0.836, against 0.063 for a random graph with the same club strengths. These communities are national markets, some closed (Italy, Turkey, Poland) and most merged into regional blocs that follow language and history (Brazil–Portugal, France–Benelux, post-Soviet, former Yugoslav, Scandinavian). Centrality has several dimensions: Italian and Brazilian loan systems lead by volume, Balkan and Eastern Mediterranean clubs by closeness, and clubs such as Benfica, Sporting CP and Red Bull Salzburg bridge the blocs. PageRank identifies the modest clubs where player paths end rather than prestigious clubs, because players are loaned down and bought up the economic hierarchy. The results are robust to the main modeling choices. Finally, we turn the network into a decision tool for Paris Saint-Germain, which recovers 0.36 euro in departure fees per euro its former academy players are later sold for.

1 Introduction

Every season, a very large number of players move between clubs (202,275 moves are recorded for 2023 alone in our dataset). Each move links a selling club to a buying club, so the transfer market is naturally a network: some clubs train players, others buy them, lend them or resell them, and players follow paths through this network over their careers. Network analysis [23] gives a vocabulary for questions that are hard to ask with tables alone: which clubs are hubs or brokers, how far apart two clubs are, and whether the market splits into groups of clubs that trade mainly among themselves.

Previous network studies of football transfers focused on elite clubs: Liu et al. [18] studied about 400 clubs of 24 top leagues (2011–2015) and related coreness and brokerage to sporting and financial success; Matesanz et al. [19] built a dynamic network of 21 European first divisions. We take a broader view, with about 12,000 clubs from all levels recorded in the dataset, including lower divisions and youth teams, and our research question is:

> *Which clubs occupy the most central positions in the football transfer network observed in this dataset, and which communities of clubs mainly exchange players with each other?*

The question matters for clubs (where to sell, loan or scout players), for leagues and regulators (how open or closed national markets are) and, methodologically, because classic centrality measures need to be read carefully on a flow network. We stress that we study *the network observed in this*

dataset, collected from Transfermarkt by a scraper, not the entire worldwide market. Our contributions are: (i) a justified graph model built from non-graph relational data; (ii) exact connectivity, distance and centrality computations on the full graph; (iii) a community analysis tested against a null model and alternative algorithms; (iv) sensitivity analyses on the modeling choices; and (v) an application, *Paris Talent Flow*, which uses the graph to help a club's sporting director. All code is available at `https://github.com/aarensberg/football-transfer-network-analysis`.

2 Dataset Presentation

2.1 Source

We use the *Soccer Player Transfer Dataset* published on Kaggle by Otavio Brito [6]. According to its author [5], the data were scraped from Transfermarkt [28] with custom Scrapy spiders starting from four continental entry points, and contain more than 3 million transfers, more than 12,000 clubs and 108 countries from 1870 to 2024. We chose it because it is large, recent, open, and not a graph: it is a set of relational tables (`transfers`, `clubs`, `leagues`, `club_league`, `players`), so we had to design and justify the graph model ourselves. The author does not document the coding of transfer types and fees, so we inferred it from the data.

Our inspection confirms the announced orders of magnitude: `transfers.csv` has 3,469,314 rows from 1870 to 2027 (2024 is incomplete and 186 rows are dated 2025–2027), with no missing values and no duplicates; `clubs.csv` describes 12,963 clubs; `leagues.csv` has 475 leagues in 111 countries. Three findings shaped the model:

- **Placeholder clubs.** Three identifiers are not clubs: `-5` ("Retired", 201,836 arrivals), `75` ("Unknown") and `515`, absent from `clubs.csv`, the current club of 91,412 players, which we interpret as "without club". 20.0% of rows involve one of them.
- **Undescribed clubs.** The transfers reference 75,177 club identifiers, but only 12,726 are described in `clubs.csv`. The 62,450 undescribed real clubs have no name or country and a median of 6 transfers, against 281 for described clubs.
- **Types and fees.** `transfer_type` is *Unknown* (50.6%), *Not loan* (39.5%), *Loan* (6.6%) or *End of loan* (3.3%); the share of *Unknown* falls from about 90% in the 1950s to 40% in the 2020s. Fees are in euros (the maximum, 222 million, is Neymar's move from Barcelona to Paris Saint-Germain in 2017), but only 1.7% of rows have a positive fee: a zero means a free transfer *or* an undisclosed fee, and the two cannot be told apart.

Joining `club_league` on (club, season) requires care: 3,215 pairs appear twice (Apertura/Clausura, two divisions), which would duplicate rows.

2.2 Graph model

Nodes and edges. Nodes are clubs; a directed edge $u \to v$ exists if at least one player moved from club u (departure) to club v (arrival). Direction matters because selling and buying are different roles (training clubs versus buying clubs). We aggregate the transfer multigraph into a *simple weighted* graph (`networkx.DiGraph`): the edge weight is the number of transfers from u to v. This keeps the multiplicity information while remaining compatible with the algorithms we need (PageRank, Louvain, shortest paths), and avoids multigraphs, which some NetworkX algorithms we use do not support (eigenvector centrality, clustering). Fees are not used as weights because 98% of them are zero or unknown.

Attributes. Nodes carry `name`, `country` and `league` (league of the most recent season up to 2023, one row per club to avoid duplication), `current_mv` (current market value) and `youth_or_reserve`, a flag from a name-based heuristic (`U19`, `Under 17`, `Primavera`, a trailing `B`/`II`/`2`, etc.). We checked it manually and corrected false positives such as Boca Juniors or Willem II. Edges carry `weight`, `total_fee` and the counts of permanent, loan and unknown transfers.

Filters. Each filter is justified by the inspection (Table 1): (1) seasons 2010–2023: complete seasons, a homogeneous contemporary period with a lower share of *Unknown* types and more stable coverage; (2) *End of loan* rows are excluded, because a return from loan is the mechanical counterpart of the loan, not

a new decision, and would duplicate every loan with a reverse edge; (3) placeholder clubs are excluded, since keeping them would create artificial hubs ("without club" would be the highest-degree node) and fictitious paths; (4) both clubs must be described in `clubs.csv`, because centralities and communities cannot be interpreted without a name and a country; (5) self-loops (8 in the whole file) are removed. Youth and reserve teams are kept, since the data contain no reliable link to the parent club; we test their removal in Section 4.7.

Table 1: Filtering funnel from raw rows to graph edges.

Step	Rows left
Raw transfer rows	3,469,314
Seasons 2010–2023	2,237,411
Without *End of loan*	2,150,643
Without placeholder clubs (`-5`, `75`, `515`)	1,722,261
Both clubs described in `clubs.csv`	920,310
Without self-loops (= graph weight)	920,304

The most costly filter is the restriction to described clubs (−802,000 rows): the network studied is the network of clubs in the leagues covered by the scraper. Coverage is also uneven: Germany has 2,238 clubs, far ahead of Italy (769) and Austria (688), because Transfermarkt covers German-speaking amateur leagues in great depth.

2.3 Graph statistics

The resulting graph is **directed, simple, weighted and without self-loops**, with **12,009 nodes and 450,217 edges** (Table 2). The heaviest edges are internal promotions between youth teams of the same club (Hertha BSC U17 → U19: 225 transfers; Bayern Munich U17 → U19: 208). Youth and reserve teams represent 1,640 nodes and 17.3% of the total weight.

Table 2: Main statistics of the transfer graph (2010–2023).

Nodes (clubs)	12,009	Countries	110
Directed edges	450,217	Undirected edges (pairs)	357,567
Transfers (total weight)	920,304	Density	3.12×10^{-3}
Mean in-/out-degree	37.5	Max in-/out-degree	202 / 278
Mean in-/out-strength	76.6	Max in-/out-strength	442 / 926
Edge weight: median / mean / max	1 / 2.04 / 225	Edges of weight 1	64.0%
Reciprocity (edges)	0.412	Youth/reserve nodes	1,640

3 Description of the Analysis

3.1 Software and conventions

The analysis is written in Python 3.14 with NetworkX 3.7 [13], pandas [20], SciPy [29] (sparse shortest paths) and Matplotlib [16]. The whole pipeline is run by one command (`uv run ftna`, about 23 minutes); each step writes a Markdown report in `results/`, and the README explains how to reproduce it. The random seed is 42 everywhere. The main conventions are:

- **Distances** are counted in hops: weights measure exchange intensity, not length. They are computed *exactly* by a breadth-first search from every node (`scipy.sparse.csgraph`, by blocks of 500 sources). Directed distances are computed on the largest strongly connected component (SCC), where all distances are finite; undirected distances on the largest weakly connected component (WCC).
- **Undirected weighted graph** (communities, eigenvector, participation): the weight of $\{u, v\}$ is $w(u \to v) + w(v \to u)$, aggregated explicitly because `to_undirected()` does not sum the two directions.
- **Baselines**: the Erdős–Rényi graph $G(n, m)$ [10] and configuration models [21] with the same (in- and out-) degrees, which are a fairer reference for heterogeneous degrees.

3.2 Methods

Structure. Degree and strength (weighted degree [1]) distributions, reciprocity, average clustering on the undirected unweighted graph, degree assortativity [22] and k-core decomposition [26].

Connectivity. Weakly and strongly connected components; bow-tie decomposition [7] around the largest SCC (IN: clubs that reach the core but cannot be reached from it; OUT: the reverse); articulation points, bridges and biconnected components of the largest WCC, to test whether any single club holds the network together.

Distances. Exact eccentricities, diameter, radius, center and periphery, both directed (out- and in-eccentricity on the largest SCC) and undirected (largest WCC); distribution of distances; small-world coefficient $\sigma = (C/C_{\text{config}})/(L/L_{\text{config}})$ [31, 15] with exact distances in configuration models.

Centrality. We use several measures because "central" has several meanings in a flow network:

- *in/out-degree* (distinct partners) and *in/out-strength* (transfers): activity;
- *closeness* (in and out), exact on the largest SCC, and with the Wasserman–Faust correction for disconnected graphs [30] on all clubs: how few transfers separate a club from all others;
- *betweenness*, exact Brandes algorithm [3], unweighted, normalized by $(n-1)(n-2)$; we parallelized it over batches of source nodes (`betweenness_centrality_subset`), and checked that the sum equals NetworkX's value to 10^{-12} on a test graph: brokerage between sub-markets;
- *PageRank* [4] with $\alpha = 0.85$ and the transfer count as weight, where the random walk follows players from seller to buyer (destinations); *reverse PageRank* on the transposed graph (suppliers); we also tested $\alpha = 0.8$;
- *eigenvector centrality* on the largest WCC (undirected, weighted), since the directed graph is not strongly connected.

We compare the rankings with Spearman correlations and validate them externally against the clubs' mean squad market value over 2010–2023 (`club_league`, 5,182 clubs with a positive value) and their current value.

Economic hierarchy. For each transfer whose two clubs have a positive market value in the season of the transfer, we compute $\log_{10}$(buyer value/seller value) by category (loan, paid permanent, permanent without fee, unknown type), using the maximum value per (club, season) to avoid duplicated rows.

Communities. The main method is **Louvain** [2] (modularity [24], resolution $\gamma = 1$, seed 42) on the undirected weighted graph. Its robustness is tested with 10 seeds, the resolutions $\gamma \in \{0.5, 1, 2, 4\}$, **Leiden** [27], Louvain on the directed graph (directed modularity [17]) and asynchronous label propagation [25], which optimizes no objective function. Its significance is tested with Louvain on 3 realizations of the weighted configuration model (club strengths preserved, transfers paired at random), which is the null model of modularity. Partitions are compared with the normalized mutual information (NMI) [8] and the adjusted Rand index (ARI) [14], coded by hand and checked on test cases. We relate communities to countries (national purity, NMI) and inspect the levels of the Louvain hierarchy. We also aggregate the graph by country to measure domestic shares, net balances and corridors.

Roles. Following Guimerà and Amaral [12], each club gets a participation coefficient $P_i = 1 - \sum_s (k_{is}/k_i)^2$ (how evenly its transfers are spread across communities, weighted) and a within-community z-score (how strongly it is connected inside its community); hubs have $z \geq 2.5$, and the roles R1–R7 follow the original thresholds. We also compute the exact edge betweenness, to check whether the edges on most shortest paths link communities.

Sensitivity. We rebuild the graph without youth/reserve teams, for 2010–2016 and 2017–2023 separately, without *Unknown* transfers and with *End of loan* rows, and compare structure, communities (NMI) and centralities (Spearman, top-50 overlap). In these variants, betweenness is approximated with 1,000 sampled sources (Spearman 0.964 with the exact value on the main graph).

3.3 Application: Paris Talent Flow

To show how the network can support a decision, we built a tool for the sporting director of Paris Saint-Germain (PSG). We listed ten problems of a club's recruitment department and scored them on added value and feasibility with our data; the best one was: *where do the players trained at PSG go, who captures the value they create afterwards, and through which springboard clubs should they be sent to recover part of it?* The tool combines careers, the graph and the community roles:

- **Careers** are rebuilt from the raw `transfers.csv`, not from the filtered graph, because players enter academies through club identifiers missing from `clubs.csv`. A *trained player* is one whose first move into a youth or reserve team of the academy happened between 2008 and 2019. We compare 15 academies, with youth teams matched manually.
- A *departure* is the first non-loan move out of the club (first team and youth teams); departures up to 2021 are evaluated. The *later value* is the highest non-loan fee later paid for the player, excluding moves involving the academy, until a possible return to the academy (after a buy-back, the career belongs to a new departure). Rows within a year are not in chronological order in the file, so all computations work at the year level.
- The **capture ratio** is $\sum$ departure fees / $\sum$ highest later fees, with a 95% bootstrap interval over players (1,000 resamples) [9]. A ratio below 1 means that most of the value was realized elsewhere.
- The *elite* is the 20 clubs with the highest squad value in each season. A player "reached the elite" if the player joined an elite club (loans included) within three years of his departure. Springboard rates use Wilson intervals [32].
- Graph tools: personalized PageRank from the PSG teams, the Louvain communities and the participation coefficient P of the first host club.

3.4 Reused code and use of AI tools

Reused code. No existing code for this dataset was reused: in particular, we did not use the author's scraper, only the published CSV files. All analysis code was written by us and relies only on the open-source libraries cited above (NetworkX, pandas, SciPy, Matplotlib); the unit tests use pytest.

Use of AI tools. We used an AI assistant (Claude Code, by Anthropic) to help write and review the analysis code, and to draft and proofread the README and this report. The research question, the modeling choices and the conclusions were decided or validated by the authors, and every number in the report comes from the outputs of the code in the repository.

4 Results

4.1 Global structure

There are **no scale-free hubs**: the complementary cumulative distributions of degree and strength decay smoothly and are cut off beyond about 100 partners or 200 transfers (99th percentile of out-degree: 145; maximum 278). A club's activity is bounded by its squad size, so degree discriminates little at the top. Clubs that buy a lot also sell a lot (Spearman 0.908 between in- and out-degree). The network is strongly **reciprocal** (41.2% of edges, against 0.3% expected in $G(n, m)$; 55.2% of transfers lie on reciprocal pairs) and **locally cohesive**: the average clustering is 0.336, against 0.015 in a degree-preserving configuration model (22 times higher). Degree assortativity is strongly positive (0.490 from seller to buyer, 0.501 undirected): active clubs trade with active clubs, as in social networks. The k-core decomposition reaches $k = 76$; surprisingly, the main core is not the European elite but the Turkish professional pyramid (139 clubs, e.g. Gençlerbirliği, Adana Demirspor), and the 60-core (8.7% of clubs) is dominated by Italy (320), Turkey (211), England (144) and Brazil (99).

4.2 Connectivity

The graph is **neither weakly nor strongly connected** (Table 3). It has 6 WCCs: the largest holds 11,999 clubs (99.92%) and the five others are isolated pairs (Japanese university teams, an Estonian

U19). It has 404 SCCs: the largest holds 11,598 clubs (96.58%) and 99.65% of transfers, followed by one SCC of 3 nodes, six of 2 and 396 singletons. Around this core, the IN part (259 clubs) is largely made of youth teams (45%) and Japanese and Estonian clubs that only supply players, and the OUT part (138 clubs) contains clubs that did not send players back during the period. A player can therefore move from almost any club to almost any other: the largest SCC is the natural component for directed distances.

Table 3: Bow-tie decomposition around the largest strongly connected component.

Part	Clubs	Share	Mean degree	Youth share	Main countries
Largest SCC	11,598	96.6%	77.5	12.9%	Germany, Italy, Austria
IN	259	2.2%	3.96	45.2%	Japan, Estonia, Germany
OUT	138	1.1%	2.51	16.7%	Japan, Germany, Turkey
Others	14	0.1%	1.71	28.6%	Japan, Estonia, France

Is any club critical for cohesion? The largest WCC has 180 articulation points and 181 bridges, so its vertex and edge connectivity are both 1. However, its largest biconnected component contains 98.46% of its clubs, and removing any single club detaches at most 4 others (Japanese amateur clubs, Jarota Jarocin, FA Tartu Kalev). **No club is critical**: cohesion is distributed.

4.3 Distances

Table 4 reports exact distances. On the largest SCC (directed), the **diameter is 12 and the radius 6** (same values for in- and out-eccentricity); on the largest WCC (undirected), the diameter is 10 and the radius 5. Distances are short: 80% of directed pairs are 3 or 4 transfers apart (Figure 1), and the mean distance (3.65 directed, 3.28 undirected) is close to that of a degree-preserving random graph (2.95 and 2.74), while clustering is 22 times higher. The network is a **small world** with $\sigma = 18.7$. The naive Erdős–Rényi estimate $\ln n / \ln\langle k\rangle \approx 2.30$ assumes homogeneous degrees and is too optimistic.

Table 4: Exact distances (in hops) and configuration-model references.

Graph	Nodes	Diameter	Radius	Mean distance	Clustering
Directed, largest SCC	11,598	12	6	3.652	
configuration model	11,598	6	4	2.953	
Undirected, largest WCC	11,999	10	5	3.281	0.336
configuration model	11,999	6	4	2.736	0.015

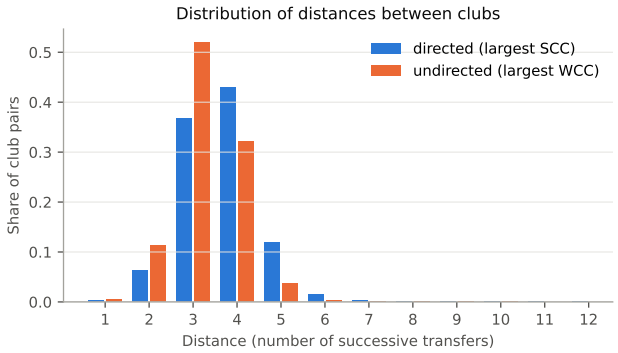


Figure 1: Share of club pairs at each distance (exact, in hops).

Diameter and radius are worst-case measures driven by the most isolated periphery: the directed periphery is made of Japanese amateur clubs (Eveil FC Kobe, Nakaniida SC) and a French U19 team, and the center contains clubs of no particular importance (PSG B, Lille B, Flora Tallinn; 418 clubs are tied at the in-radius). They describe the reach of the network, not which clubs matter; closeness, an average over all distances, is more informative.

4.4 Centrality

Table 5 shows the top clubs of each measure. Rank correlations between distinct measures range from 0.41 to 0.94 (the closeness variants aside), and each measure points to a different kind of club, so **centrality is multidimensional**:

- **Volume** (strength, degree): Italian and Brazilian clubs, whose loan systems create very high activity. Parma made 926 outgoing transfers to 278 clubs (a genuine outlier, not an artifact: its loan and co-ownership system before its 2015 bankruptcy). The top 100 by in-strength is 43% Italian and 31% Brazilian.
- **Closeness**: international hubs of the Balkans and the Eastern Mediterranean (Hajduk Split, Rijeka, Dinamo Zagreb, Red Star, Partizan, Olympiacos for out-closeness; Cypriot, Romanian and Greek clubs for in-closeness). The Wasserman–Faust variant gives the same ranking (Spearman 1.000 on the SCC), so the choice of component does not bias it.
- **Betweenness**: mid-level clubs that link sub-markets, especially between the very dense German amateur network (a coverage effect) and the rest: Rehden, Uerdingen, Neuhof, Mosta (Malta), Spartaks Jūrmala (Latvia). Parma ranks 14th.
- **Reverse PageRank** (suppliers): the top 20 are all youth academies (Red Bull Salzburg U16, Rapid Wien U16, Ajax U19).
- **PageRank** (destinations): modest clubs such as Kalev Tallinn, FSV Duisburg or Finnish clubs, where many player paths end.
- **Eigenvector**: its top 100 is entirely Brazilian (58) and Italian (42): the measure localizes on the two densest loan sub-networks. This is the known weakness that PageRank's teleportation corrects, which is why we rely on PageRank rather than eigenvector centrality for global importance in the directed flow. Changing α from 0.85 to 0.8 does not matter (Spearman 0.999).

Table 5: Top 5 clubs for each centrality measure, and main countries of the top 100.

Measure	Top 5	Top-100 countries
Out-strength	Parma (926), Atalanta (718), Genoa (717), Tombense (650), Inter (582)	Italy 34, Brazil 21, Turkey 10
In-strength	Parma (442), Ponte Preta, Bragantino (389), Guarani, ABC (374)	Italy 43, Brazil 31, Turkey 16
Out-closeness	Hajduk Split, Rijeka, Udinese, Istra 1961, Dinamo Zagreb	Croatia 13, Greece 11, Italy 9
In-closeness	Mosta, Sheriff Tiraspol, Istra 1961, Botoșani, Ermis Aradippou	Cyprus 11, Turkey 9, Romania 8
Betweenness	Rehden, Uerdingen, Neuhof, Mosta, Spartaks Jūrmala	Germany 35, Finland 15, Austria 8
PageRank	Kalev Tallinn, FSV Duisburg, PK-35, FC Kiffen, Tampere United	Germany 29, Switzerland 10, Finland 9
Reverse PageRank	Salzburg U16, Erfurt U17, Rapid Wien U16, Sarajevo U17, Ried U16	Germany 37, Austria 14, Italy 10
Eigenvector	Parma, Tombense, Genoa, Mirassol, Ponte Preta	Brazil 58, Italy 42

Which measure captures economic importance? Table 6 correlates each measure with the clubs' mean market value over 2010–2023. **Out-closeness is the most aligned** (0.72): economically strong clubs can send players anywhere in few steps. **PageRank is almost unrelated** (0.08). Figure 2 explains why: elite clubs and academies lie below the diagonal (strong suppliers, weak destinations), while modest clubs lie above it. In this network, PageRank, designed as a prestige measure for the web, measures where player paths end (modest, often lower-tier clubs), not prestige.

Economic hierarchy. The direction of transfers confirms this reading (Figure 3). Among the 324,369 transfers whose two clubs have a market value (35.2% of the graph), 61.3% go to a less valuable club. **85.4% of loans go down** the hierarchy (median buyer about 3 times less valuable), while **64.5% of**

Table 6: Spearman correlation between selected centralities and club market value.

	Out-clos.	Out-deg.	Out-str.	In-clos.	In-deg.	Rev. PR	Betw.	PageRank
Mean value 2010–2023	**0.72**	0.66	0.63	0.61	0.59	0.53	0.43	0.08
Current value	0.59	0.56	0.54	0.53	0.53	0.47	0.36	0.03

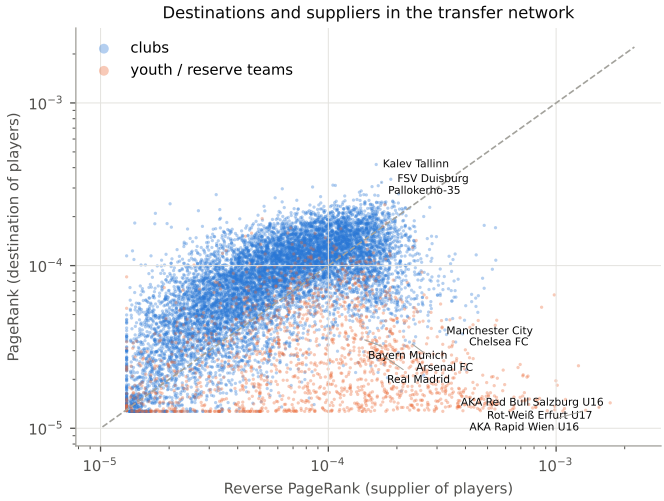


Figure 2: PageRank (destinations) against reverse PageRank (suppliers), log scales. The five clubs with the highest current market value and the youth academies lie below the diagonal.

paid transfers go up (median buyer 1.6 times more valuable). Players are loaned down and bought up; the values of the previous season change these shares by about 3 points at most. Rich clubs are upstream of the flows, which explains their high reverse PageRank and low PageRank.

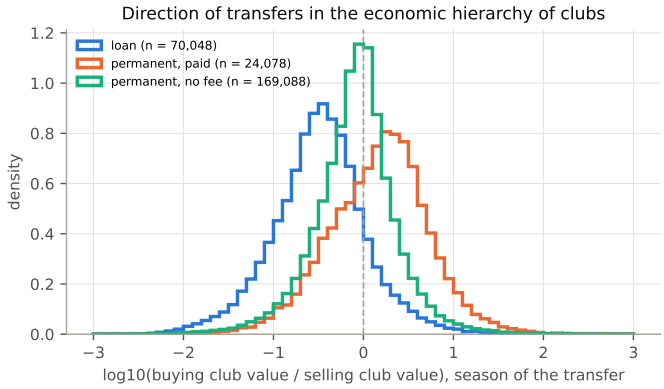


Figure 3: Ratio of buyer to seller market value (season of the transfer), by transfer category.

4.5 Communities

Louvain finds **35 communities with a modularity of 0.836**. This is far above the weighted configuration model, where Louvain reaches only 0.063 ± 0.001 (at most 0.064 over 3 realizations, with 6 to 9 communities): the communities are not an artifact of the algorithm. The 15 largest cover 78.9% of the clubs. The partition is robust (Table 7): modularity is identical over 10 seeds (34 to 37 communities, mean NMI 0.976, ARI 0.933); Leiden (NMI 0.992) and directed Louvain (NMI 0.987) find almost the same partition; label propagation, which optimizes nothing, finds a finer but consistent partition (332 communities, NMI 0.781 with Louvain). Changing the resolution merges or splits blocs (25 communities at $\gamma = 0.5$, 88 at $\gamma = 4$) while agreement with countries stays high (NMI 0.81 to 0.90); we keep $\gamma = 1$ as the standard choice, keeping in mind the resolution limit of modularity [11].

Table 7: Comparison of partitions (modularity on the undirected weighted graph).

Partition	Communities	Modularity	NMI with Louvain	NMI with country
Louvain (reference)	35	0.836	1.000	0.849
Leiden	34	0.836	0.992	0.852
Louvain, directed modularity	35	0.836	0.987	0.853
Label propagation	332	0.771	0.781	0.871
Countries (110 + unknown)	111	0.806	0.849	1.000
Weighted configuration model	6–9	0.063	–	–

Interpretation. The country alone already gives a modularity of 0.806, and 84.9% of transfers are domestic. Louvain improves on it by **merging most national markets into regional blocs** that follow language, history and geography (Table 8, Figure 4): France–Belgium–Netherlands (C2), the post-Soviet space (C4), Spanish-speaking America (C6), East and Southeast Asia (C7), Scandinavia (C8), the British Isles (C9), the former Yugoslavia (C10), Greece–Bulgaria–Romania–Cyprus (C14), the former Czechoslovakia and Hungary (C15), the Portuguese-speaking bloc Brazil–Portugal (C18), the Maghreb and the Gulf (C20), Greater China (C21), North America (C22), sub-Saharan Africa (C23) and Australia–India–New Zealand (C24, Australian players in the Indian Super League). Other markets form closed national communities: Italy (98% Italian, 94% of its members' transfers inside the community), Austria, Turkey, Switzerland, Finland, Poland and Iran (96% internal). Germany is split in two (C13: Hamburg, Bremen and Schleswig-Holstein in the north-west; C1: the rest of the country), a regional structure that is visible only because amateur leagues are covered so finely.

Table 8: The 20 largest Louvain communities. Internal: share of members' strength inside the community.

C	Clubs	Main countries (clubs)	Internal	Most active clubs
1	1,889	Germany (1,869)	0.93	Berliner AK, Lotte
2	890	France 499, Belgium 165, Netherlands 144	0.85	Monaco, Standard Liège
3	787	Italy 769, San Marino 16	0.94	Parma, Genoa, Atalanta
4	774	Russia 219, Ukraine 140, Belarus 65	0.89	Khimki, Dinamo Tbilisi
5	705	Austria 688	0.91	Red Bull Salzburg, LASK
6	692	Argentina 136, Mexico 98, Peru 68	0.93	Deportivo Pasto, América de Cali
7	572	Japan 326, Thailand 66, South Korea 64	0.91	Muangthong United
8	532	Norway 179, Denmark 135, Sweden 114	0.89	Midtjylland, Eskilstuna
9	503	England 273, Scotland 84, Wales 75	0.92	Blackpool, Stevenage, Chelsea
10	476	Bosnia 139, Serbia 78, Slovenia 67	0.84	Lokomotiva Zagreb, Red Star
11	386	Turkey 386	0.93	Gençlerbirliği, Adana Demirspor
12	336	Switzerland 315	0.88	Chiasso, Sion
13	314	Germany 313 (north)	0.92	Concordia Hamburg, Altona 93
14	309	Greece 118, Bulgaria 92, Romania 52	0.77	Olympiacos, Ermis Aradippou
15	306	Czech Rep. 165, Slovakia 81, Hungary 58	0.85	Slavia Prague, DAC
16	301	Spain 266, Andorra 17, Gibraltar 16	0.85	Granada, Real Murcia
17	301	Finland 301	0.92	HJK Klubi 04, PK-35
18	300	Brazil 179, Portugal 121	0.84	Tombense, Cruzeiro
19	267	Poland 266	0.89	Jagiellonia, Górnik Zabrze
20	233	Tunisia 46, Saudi Arabia 46, Algeria 46	0.87	Club Africain, ES Sétif

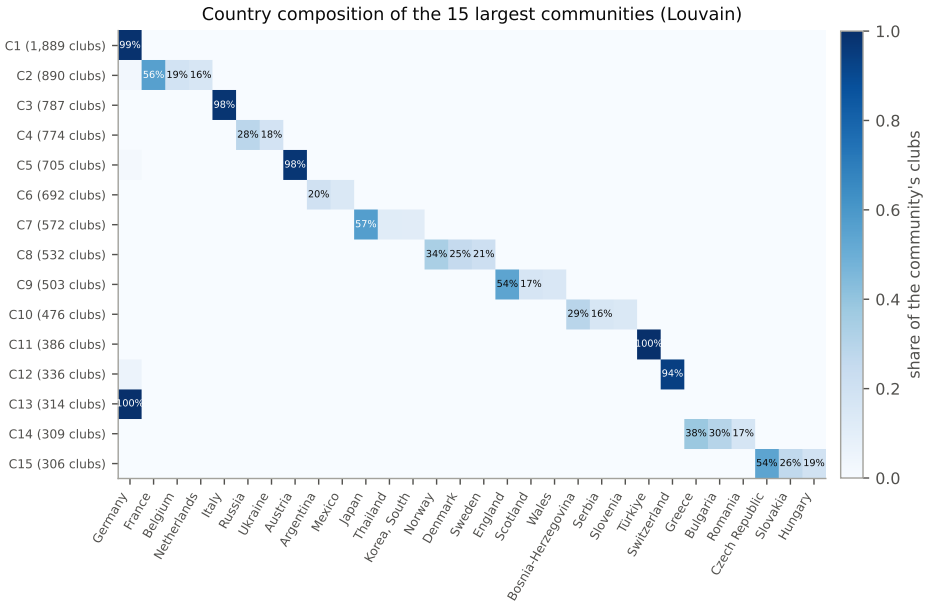


Figure 4: Share of each country in the 15 largest communities (columns ordered by community).

Nesting. The levels of the Louvain hierarchy show this nesting directly: the first level has 389 communities with a national purity of 0.98 (national and regional sub-markets); at the second level, countries merge into 52 transnational blocs (purity 0.72), and the third and fourth levels give 36 and then the final 35 communities.

Country flows. The country network confirms the blocs. Of the 138,785 international transfers, 31.2% stay within a bloc, and 16 of the 25 largest corridors link two countries of the same bloc: England → Scotland (1,332), France → Belgium (900), Brazil → Portugal (888), Argentina → Chile (685), Czech Republic → Slovakia (660). The main exceptions are Germany ↔ Austria (847 and 755): the very large German domestic volume keeps the two countries in separate communities. The most closed countries (at least 20 clubs) are Iran (93% domestic), Germany (90%), Hong Kong and Italy (89%); the most open are Lithuania (37%), Cyprus (41%), Azerbaijan (43%) and Croatia (46%). England (+2,786), France (+2,338) and Brazil (+1,427) are the largest net exporters, partly through youth loans abroad; Turkey (−1,470), Austria (−1,026) and the United States (−915) are the largest net importers.

4.6 Bridges between communities

Figure 5 contracts each community into one node. The France–Benelux bloc (C2) is the main crossroads, with strong links to Germany, Italy, Spain, the British Isles, Brazil–Portugal and Greece; the former Yugoslav bloc (C10) links Austria, Central Europe, the post-Soviet space and Greece; peripheral blocs (Iran, Finland, sub-Saharan Africa, Australia–India) have only weak links.

Which clubs connect the blocs? The participation coefficient P answers this at the club level (Table 9). Most clubs trade within one community (median P of 0.11 among the 6,506 active clubs with a strength of at least 100). Over all 12,009 clubs, the roles are 4,705 ultra-peripheral and 6,954 peripheral clubs, 183 connectors, 7 kinless clubs, 148 provincial hubs and 12 connector hubs. Two profiles bridge the blocs: **exporting "stepping-stone" clubs** (Sheriff Tiraspol, Sporting CP, Benfica, Red Bull Salzburg, Olympiacos, Dynamo Kyiv, Basel, Copenhagen) and **elite clubs that recruit and loan internationally** (Manchester City, Chelsea, Atlético de Madrid, Barcelona, Sevilla, Marseille). Hubs with a high z, on the contrary, are mostly local: youth and reserve teams, Indian and Hong Kong clubs, and Parma ($z = 5.77$). All centralities follow z, i.e. activity inside the community, most closely (Spearman 0.71 to 0.82). P adds a distinct, partial dimension: it correlates with out-closeness (0.65) and betweenness (0.60), but hardly with PageRank (0.27), so closeness and betweenness also reward bridging between blocs. Edge betweenness tells the same story: 75.5% of the 1,000 edges with the highest betweenness link two communities, against 19.7% of all edges.

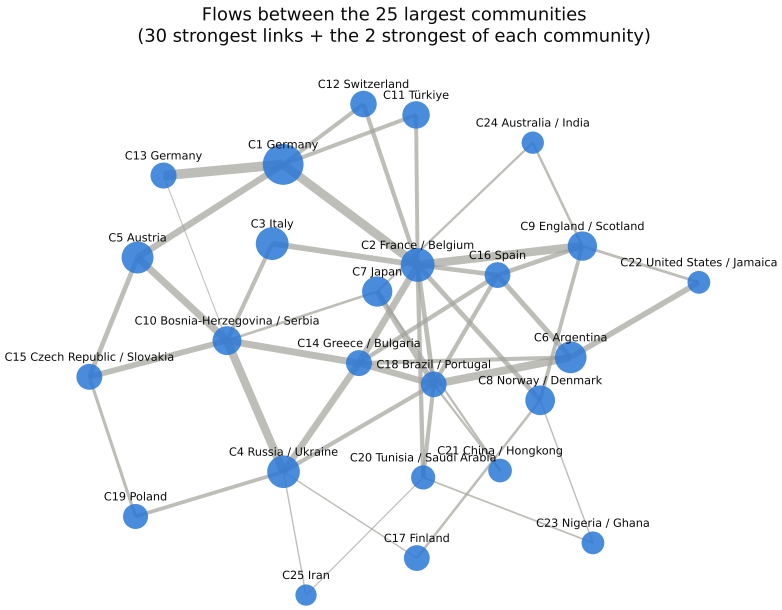


Figure 5: Graph of the 25 largest communities (node size: number of clubs; edge width: transfers between communities; the 30 strongest links and the 2 strongest links of each community are shown).

Table 9: Bridge clubs: highest participation coefficient among active clubs (strength ≥ 100).

Club	Bloc	P	z	Club	Bloc	P	z
Sheriff Tiraspol	C4	0.82	0.35	Chelsea	C9	0.78	−0.02
Sporting CP	C18	0.81	−0.06	Dynamo Kyiv	C4	0.78	0.25
Benfica	C18	0.81	0.11	FC Basel	C12	0.78	0.58
Red Bull Salzburg	C5	0.81	0.83	Panathinaikos	C14	0.78	0.33
Manchester City	C9	0.79	−0.33	Vitesse	C2	0.77	0.21
Olympiacos	C14	0.79	1.76	FC Copenhagen	C8	0.77	0.32

4.7 Robustness to modeling choices

Table 10 compares five variants with the reference graph. Structure is stable (reciprocity 0.36–0.48, clustering 0.31–0.35, assortativity 0.48–0.52, modularity 0.82–0.84). Communities are preserved (NMI $\geq$ 0.92) and centrality rankings are stable overall (Spearman $\geq$ 0.78). The exact top ranks move for all measures (top-50 overlap 0.42–0.96 outside PageRank and betweenness), and most for PageRank and betweenness (0.24–0.80 and 0.44–0.60), so we interpret *profiles* of clubs rather than exact ranks. Two variants are informative. Without youth teams, the top 10 of reverse PageRank becomes Genoa, Parma, Juventus, Inter, Chelsea, Udinese, Atalanta, Roma, Benfica and Sampdoria: the big lending clubs, consistent with the hierarchy result. Adding *End of loan* rows inflates reciprocity (0.41 to 0.48) with mechanical reverse edges and changes nothing else, which validates their exclusion. Diameters grow in the sparser sub-period graphs (15 and 13).

Table 10: Sensitivity analysis (diameter and radius on the largest SCC; Spearman and NMI computed on the clubs shared with the reference).

Variant	Nodes	Diam./rad.	Comm.	Modul.	NMI comm.	ρ out-clos.	ρ betw.
Reference	12,009	12 / 6	35	0.836	1.000	1.000	1.000
Without youth/reserve	10,357	9 / 5	31	0.829	0.953	0.998	0.926
2010–2016	10,973	15 / 10	29	0.835	0.920	0.908	0.781
2017–2023	11,380	13 / 7	39	0.837	0.965	0.940	0.830
Without *Unknown*	11,784	13 / 7	32	0.817	0.951	0.973	0.890
With *End of loan*	12,009	12 / 6	34	0.837	0.977	0.996	0.938

4.8 Application: Paris Talent Flow

We now apply the network to the question of Section 3.3: what happens to the players trained at PSG after they leave?

Value realized elsewhere. 239 players were trained at PSG (entry 2008–2019) and 215 departures can be evaluated; 91% of them have no known fee. PSG received **€118.7M** when these players left, while the highest fees later paid for them add up to **€325.3M**: the capture ratio is **0.36** (95% CI [0.23; 0.58]; 0.41 without the two largest cases). It is the lowest of the 15 academies as a point estimate, close to Rennes (0.39) and Bayern (0.40), whose intervals overlap with PSG's (Figure 6, Table 11). Ajax, Monaco, Lyon and Benfica are at 1.29–1.36, Real Madrid at 1.17 and Juventus at 2.40. Typical PSG cases are Nkunku (€19.5M to Leipzig, later sold for €60M to Chelsea), Diaby (€15M to Leverkusen, later €60M) and Coman (free to Juventus, later €21M to Bayern). The capture ratio ignores one lever, the buy-back, which is visible separately: Juventus bought back 16 of its former academy players, PSG 9 and Real Madrid 8.

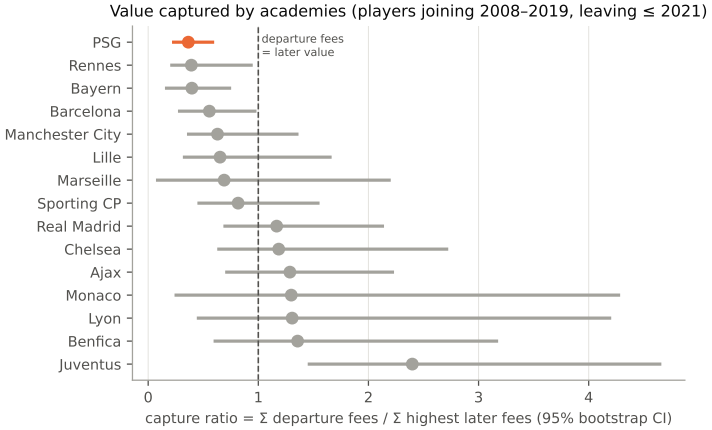


Figure 6: Capture ratio of 15 academies with 95% bootstrap intervals (players joining 2008–2019, leaving up to 2021).

Table 11: Selected academies: departure fees, highest later fees and capture ratio.

Academy	Departures	Paid	Fees (€M)	Later fees (€M)	Ratio [95% CI]	Bought back
PSG	215	20	118.7	325.3	0.36 [0.23; 0.58]	9
Rennes	103	19	133.0	339.3	0.39 [0.21; 0.93]	2
Bayern	274	21	36.5	91.9	0.40 [0.17; 0.74]	3
Barcelona	141	40	152.2	274.0	0.56 [0.29; 0.97]	6
Manchester City	154	36	197.9	314.3	0.63 [0.37; 1.35]	1
Lille	134	15	86.2	132.0	0.65 [0.33; 1.65]	3
Real Madrid	151	37	316.5	271.2	1.17 [0.70; 2.13]	8
Ajax	223	40	398.6	309.7	1.29 [0.71; 2.22]	4
Lyon	137	30	223.3	170.8	1.31 [0.46; 4.19]	4
Benfica	170	28	438.1	322.9	1.36 [0.61; 3.16]	5
Juventus	223	76	238.1	99.3	2.40 [1.46; 4.65]	16

Who captures the value? Weighted by later value, the first host clubs of PSG alumni are RB Leipzig (2 players, €81M of later fees), Lille (5 players, €73.8M) and Leverkusen (€60M). For the six French academies together, Monaco (€149.5M), Dortmund (€135M) and Nice (€107M) lead. The network adds two observations. First, a PageRank personalized on the PSG teams spreads mostly locally (Lens, Orléans, Paris FC, Poissy, Créteil, Angers, Lille), but the departures to graph clubs outside PSG's community C2 (33% of 154) carry 62% of the later value: value is mostly realized outside the France–Benelux bloc. Second, the participation coefficient of the first host club matters. Among academy departures to non-elite clubs (all 15 academies), players sent to a **bridge club** ($P \geq 0.6$) reached the elite within three years in 5.5% of cases, against 1.2% otherwise, and were later sold for a fee in 47.7% of cases, against 18.0%. This is an association, not a causal effect: the best young players may be sent to these clubs in the first place.

Springboards. Among 687,113 permanent exits (*Not loan* or *Unknown*) from non-elite first teams (2010–2020, youth and reserve teams excluded), only 0.17% reach the elite within two moves and three years, and only 5% of the 7,368 clubs with at least 30 exits sent any player there (median rate among them: 1.3%). A few clubs stand out (rate and Wilson 95% interval): Southampton 20.7% [13.4; 30.7], Ajax 19.8%, Lyon 20.0%, Lille 18.3%, Sevilla 18.5% and Villarreal 17.9%, then Mönchengladbach, Benfica, Genoa and Porto. The French springboards are Lyon, Lille, Marseille, Monaco, Bordeaux and Toulouse. Rising springboards (rate 2016–2020 against 2010–2015, at least 15 exits per period) include Southampton (15% to 31%), Mönchengladbach, Leicester, Betis, Flamengo and Red Bull Salzburg. With an elite of 30 clubs instead of 20, rates keep a Spearman correlation of 0.83.

Competitors and comparables. Foreign clubs signing players from French youth teams (loans excluded) are mainly Belgian (165) and Italian (123); Italian signings rose from 29 in 2010–2016 to 94 in 2017–2023. For contract clauses, we tabulate, by position and age at departure, the share of players later sold for a fee and the distribution of the highest later fee: the share ranges from 4–5% (goalkeepers aged 19 or less, or 24 and more) to 32% (forwards aged 22–23), and the 90th percentile among resold players reaches €30–33M for defenders aged 20–21 and forwards aged 19 or less. These figures can anchor sell-on percentages and buy-back prices.

Recommendations and limits. For PSG, the data suggest (i) selling academy players with sell-on clauses rather than outright, (ii) using buy-back clauses more systematically (Juventus bought back 16 former academy players), and (iii) loaning young players to measured springboards and bridge clubs (Lille, Lyon, Ajax, Benfica, Salzburg). These conclusions have clear limits: there is no performance data (minutes, statistics); a zero fee may be free or undisclosed and clauses are invisible; only cohorts that left up to 2021 can be evaluated; a dozen players carry most of an academy's value, hence the wide intervals; and youth teams are matched manually.

5 Conclusions

We modeled 920,304 transfers (2010–2023) as a directed, simple, weighted graph of 12,009 clubs and 450,217 edges, with every filter justified by a data inspection. The graph is not connected (6 WCCs,

404 SCCs), but 96.6% of clubs belong to one strongly connected component, where the diameter is 12 and the radius 6 (10 and 5 on the largest WCC, undirected). It is a reciprocal, assortative small world without scale-free hubs, and no single club holds it together.

Central clubs. There is no single center. Italian and Brazilian loan systems lead by volume (Parma, Genoa, Atalanta, Tombense); Balkan and Eastern Mediterranean clubs lead closeness, the measure most aligned with economic weight (Spearman 0.72); stepping-stone clubs (Benfica, Sporting CP, Salzburg, Olympiacos, Dynamo Kyiv, Basel) and internationally active elite clubs (Manchester City, Chelsea, Barcelona, Atlético) bridge the blocs; academies are the main sources. PageRank must not be read as prestige here: because players are loaned down (85% of loans) and bought up (65% of paid transfers), it points to the modest clubs where player paths end.

Communities. The clubs mainly exchange players within national markets (85% of transfers), grouped into 35 communities: a few closed national markets and regional blocs that follow language and history (modularity 0.836 against 0.063 at random). The communities are robust to the algorithm, the seed, edge direction, youth teams, period and transfer types. France and the Benelux form the main crossroads between blocs.

Limitations. Only clubs described in `clubs.csv` are kept, and Transfermarkt covers some countries (German-speaking amateur leagues) much more finely than others, which affects degree, betweenness and the split of Germany; 98% of fees are zero or undisclosed; youth teams are detected with a name-based heuristic; market values are available for only 35% of transfers; the exact top ranks of PageRank and betweenness depend on modeling choices.

Future work. A temporal analysis (one graph per season) could follow the rise of springboard clubs and of new markets such as Saudi Arabia; a player–club bipartite model could follow careers instead of aggregated flows; and performance data (minutes played) would allow the Paris Talent Flow tool to estimate the effect of a loan destination, not only associations.

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